

NEUBRANDENBURG, Germany

WMO: 102810

Lat: 53.5997N

Lon: 13.3039E

Elev: 228

StdP: 14.58

Time Zone: 1.00 (EUC)

Period: 95-13

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	10.8	16.1	6.6	7.8	13.3	12.2	10.3	17.1	29.2	41.3	25.7	41.0	6.7	90	0.580

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.2	84.5	66.2	81.3	65.3	77.8	64.0	68.6	79.9	66.9	77.5	65.3	75.0	8.0	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
64.7	92.2	72.1	63.0	86.8	70.7	61.5	82.2	69.5	32.9	79.9	31.5	77.5	30.3	75.1	76.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.1	20.0	17.6	5.5	91.7	7.5	3.9	0.1	94.5	-4.2	96.8	-8.4	99.0	-13.9	101.8
			5.9	72.4	7.1	2.0	0.7	73.8	-3.4	75.0	-7.4	76.1	-12.5	77.5
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	48.7	32.4	34.3	39.3	48.0	55.3	60.9	65.4	65.5	58.3	49.3	40.5	33.9
	DBStd	13.72	8.75	8.56	7.09	7.18	6.38	5.63	5.46	4.96	5.49	6.53	6.64	8.30
	HDD50	2350	546	440	337	121	20	0	0	0	3	93	289	501
	HDD65	6148	1010	859	796	513	308	148	62	55	211	487	734	965
	CDD50	1865	0	1	6	60	184	327	478	481	252	71	5	0
	CDD65	189	0	0	0	2	7	25	75	71	9	0	0	0
	CDH74	1684	0	0	0	22	102	261	674	542	82	1	0	0
	CDH80	424	0	0	0	3	18	67	198	129	9	0	0	0
(14) Wind	WSAvg	8.6	9.2	10.1	9.7	8.9	8.2	8.2	8.0	7.6	7.9	8.3	8.3	8.7
(15) Precipitation	PrecAvg	22.4	1.7	1.3	1.5	1.3	2.0	2.5	2.6	2.4	1.9	1.7	1.7	1.8
	PrecMax	30.1	3.1	2.9	3.2	2.6	4.2	5.7	9.0	6.3	5.7	3.1	3.7	3.4
	PrecMin	15.7	0.0	0.2	0.4	0.1	0.4	0.7	0.4	0.9	0.6	0.6	0.1	0.4
	PrecStd	3.1	0.9	0.6	0.7	0.6	0.9	1.1	1.7	1.3	1.1	0.7	0.7	0.7
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.3	54.6	63.0	77.0	82.3	86.6	90.2	87.7	80.8	68.4	57.5	53.0
		MCWB	48.4	48.8	52.4	58.7	64.3	67.5	69.1	66.8	63.4	59.7	52.9	48.5
	2%	DB	49.1	51.5	57.4	69.4	76.5	80.7	85.1	83.5	76.1	64.5	54.8	49.4
		MCWB	45.9	47.2	48.6	56.0	62.1	65.7	67.3	65.5	62.4	57.2	50.7	46.4
	5%	DB	46.3	48.5	53.4	64.7	71.8	76.3	81.4	80.3	71.9	61.1	51.9	46.5
		MCWB	43.7	45.0	47.5	53.4	59.4	63.3	66.0	64.9	60.5	55.5	48.8	44.1
	10%	DB	43.6	46.1	49.8	60.7	68.1	72.4	77.2	76.4	67.9	58.5	49.6	44.1
		MCWB	41.4	42.7	44.7	51.2	57.3	61.3	64.2	63.8	59.3	54.2	46.9	42.0
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	49.2	50.2	53.0	61.3	66.7	69.8	71.5	70.5	65.7	60.9	54.1	49.5
		MCDB	51.9	52.7	60.7	72.8	78.5	80.1	85.1	82.1	76.0	67.9	56.8	51.5
	2%	WB	46.1	47.7	50.3	57.1	63.5	66.9	69.0	67.9	63.7	58.0	51.3	46.5
		MCDB	48.7	50.8	55.7	68.4	74.1	78.1	80.6	78.2	72.7	62.8	53.9	48.9
	5%	WB	43.7	45.4	47.5	54.4	60.7	64.5	67.1	66.2	61.8	56.2	49.0	44.1
		MCDB	45.9	48.3	52.7	62.9	70.1	73.9	78.3	75.5	69.4	60.6	51.5	46.2
	10%	WB	41.5	42.6	45.0	51.8	58.3	62.3	65.4	64.8	60.1	54.3	47.1	41.9
		MCDB	43.6	45.7	49.2	59.6	66.4	70.9	75.4	73.8	67.0	58.2	49.3	43.7
(35) Mean Daily Temperature Range	5% DB	MDBR	8.3	9.7	12.8	17.3	18.7	17.6	17.2	17.3	15.2	12.1	8.5	7.4
		MCDBR	9.4	10.4	18.4	23.8	25.4	24.2	24.7	24.0	21.3	15.1	10.0	8.1
		MCWBR	8.0	7.8	11.7	13.1	13.6	12.3	10.3	10.3	10.4	9.4	7.6	7.2
	5% WB	MCDBR	8.9	9.9	17.0	21.9	23.1	22.1	22.0	20.1	19.1	14.4	9.5	7.9
		MCWBR	8.0	7.9	11.5	12.5	13.0	12.2	10.1	9.8	10.1	9.3	7.8	7.2
(40) Clear-Sky Solar Irradiance	taub		0.313	0.338	0.371	0.403	0.400	0.397	0.417	0.413	0.378	0.360	0.336	0.307
	taud		2.316	2.309	2.254	2.229	2.279	2.310	2.275	2.297	2.372	2.387	2.367	2.314
	Ebn at Noon		205	238	254	261	269	271	262	256	251	227	195	183
	Edh at Noon		19	26	34	39	39	39	39	36	30	24	18	16
(44) All-Sky Solar Radiation	RadAvg		187	395	773	1295	1621	1710	1592	1338	977	524	226	128
	RadStd		22	56	87	180	193	134	219	108	103	70	25	11

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	+1.85	N/A	N/A	N/A	N/A
(47) Regional (41 neighbors)	+0.68	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page